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Patent Public Search | Text View

United States Patent Application Publication

20250279381

Kind Code

A1

Publication Date

September 04, 2025

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SEMICONDUCTOR DIE HAVING A DIE LEVEL DISTRIBUTION (DLD) METALLIZATION STRUCTURE INCLUDING A METAL PAD AND A SHORTER VIA COUPLING THE METAL PAD TO A METAL INTERCONNECT IN THE DIE FOR IMPROVED SIGNAL PATH CONDUCTIVITY

Abstract

Aspects disclosed in the detailed description include a semiconductor die having a die level distribution (DLD) metallization structure including a metal pad and a short via coupled to the metal pad to a metal interconnect in the die for improved signal path conductivity. The DLD metallization structure includes an outer metallization layer comprising the metal interconnect, a first passivation layer adjacent to the outer metallization layer and a DLD metallization layer adjacent to the first passivation layer. The DLD metallization layer comprises a first surface adjacent to the first passivation layer and a metal pad. The DLD metallization structure includes a first via extending in the second direction orthogonal to the first direction, the first via coupling the metal pad and the metal interconnect, the first via having a first aspect ratio between 0.06-0.5, inclusively.

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Family ID: 96880593

Appl. No.: 18/591820

Filed: February 29, 2024

Publication Classification

Int. Cl.: H01L23/00 (20060101); H01L23/48 (20060101); H01L25/10 (20060101)

CPC **H01L24/16** (20130101); **H01L23/481** (20130101); **H01L24/32** (20130101); **H01L24/73** (20130101); **H01L25/105** (20130101); H01L24/75 (20130101); H01L24/81 (20130101); H01L2224/16225 (20130101); H01L2224/32225 (20130101); H01L2224/73204 (20130101); H01L2224/75702 (20130101); H01L2224/81203 (20130101); H01L2224/81801 (20130101); H01L2924/1436 (20130101)

Background/Summary

BACKGROUND

I. Field of the Disclosure

[0001] The field of the disclosure relates to integrated circuit (IC) packages that include one or more semiconductor dies (“dies”) attached to a package substrate, and more particularly to a die level distribution (DLD) metallization structure on a die(s).

II. Background

[0002] Integrated circuits (ICs) are the cornerstone of electronic devices. ICs are packaged in an IC package, also called a “semiconductor package” or “chip package.” The IC package includes one or more semiconductor dice (“dies” or “dice”) as an IC(s) that is mounted on and electrically coupled to a package substrate to provide physical support and an electrical interface to the die(s). The package substrate also includes one or more metallization layers that include metal interconnects (e.g., metal traces, metal lines) with vertical interconnect accesses (vias) coupling the metal interconnects together between adjacent metallization layers to provide electrical interfaces between the die(s). The package substrate also includes an outer metallization layer that includes metal interconnects coupled to external metal interconnects (e.g., ball grid array (BGA) interconnects) to provide an external interface between the die(s) in the IC package. The external metal interconnects can also be coupled (e.g., soldered) to traces in a printed circuit board (PCB) to attach the package to the PCB and interface its die(s) with the circuitry of the PCB.

[0003] The die(s) also includes one or more metallization layers that include metal interconnects (e.g., metal traces, metal lines). The one or more metallization layers are fabricated in the die(s) utilizing a back end of line (BEOL) process. A die level distribution (DLD) metallization layer includes metal interconnects and metal pads. The DLD metallization layer couples to an outer metallization layer which includes metal interconnects fabricated during the BEOL process. The die(s) also includes die interconnects (e.g., balls or pillars) which are supported by metal pads in the DLD metallization layer and electrically couple the metal interconnects in the die(s) to the metal interconnects exposed in the outer metallization layer of the package substrate or another die.

SUMMARY

[0004] Aspects disclosed in the detailed description include a semiconductor die having a die level distribution (DLD) metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity. Related integrated circuit (IC) packages and fabrication methods are also disclosed. The die includes a semiconductor layer, a DLD metallization structure, and a back end of line (BEOL) interconnect structure between the semiconductor layer and the DLD metallization structure. The DLD metallization structure mechanically supports die interconnects for coupling the die to another device, such as a package substrate or another die, and redistributes signals (e.g., power, ground, information) between the die interconnects and the semiconductor layer through the BEOL interconnect structure.

[0005] In this regard, the DLD metallization structure includes a DLD metallization layer and a

passivation layer disposed between the DLD metallization layer and an outer metallization layer in the BEOL interconnect structure, forming a diffusion barrier therebetween. Vias are formed in the passivation layer to couple metal pads in the DLD metallization layer to metal interconnects in a BEOL interconnect layer in the BEOL interconnect structure. It is desired to reduce the signal path resistance or, conversely, improve signal path conductivity between the die interconnects and the BEOL interconnect structure.

[0006] In this regard, in exemplary aspects, the passivation layer between the DLD metallization structure and the BEOL interconnect structure is provided as a thinner layer to minimize the distance between the DLD metallization structure and the BEOL interconnect structure. For example, the height or thickness of the passivation layer may be between 0.1 micrometer (μm) and 1 μm , inclusively, and is fabricated at this thin layer because the passivation layer's purpose is to merely act as a diffusion barrier.

[0007] In another exemplary aspect, a second passivation layer is provided between the DLD metallization structure and the die interconnects to act as a diffusion barrier between metal pads and traces in the DLD metallization structure. If the die is bonded to a package substrate as opposed to another die, for example, which requires a smooth coupling surface between dies, the second passivation layer can also be thinner. In a die-to-die bonding approach, a polishing step, such as a chemical mechanical polishing (CMP), is needed to fabricate the smooth coupling surface and requires a thick passivation layer to include margin material which is removed when performing the polishing step. Avoiding the polishing step enables the passivation layer to be formed as a thin layer since the passivation layer does not need to include margin material and advantageously simplifies the process of fabricating the die.

[0008] In this regard in one aspect, a semiconductor die (die), comprises a semiconductor layer extending in a first direction, a die level distribution (DLD) metallization structure and a back end of line (BEOL) interconnect structure between the semiconductor layer and the DLD metallization structure. The BEOL interconnect structure extending in a second direction orthogonal to the first direction. The DLD metallization structure comprises an outer metallization layer extending in the first direction, the outer metallization layer comprising a metal interconnect. The DLD metallization structure further comprises a first passivation layer extending in the first direction adjacent to the outer metallization layer and a DLD metallization layer extending in the first direction. The DLD metallization layer comprises a first surface adjacent to the first passivation layer and a metal pad. The DLD metallization structure further comprises a first via extending in the second direction orthogonal to the first direction, the first via coupling the metal pad and the metal interconnect, the first via having a first aspect ratio between 0.06-0.5, inclusively.

[0009] In another aspect, a method of fabricating a semiconductor die (die) for improved signal path conductivity, comprises fabricating a semiconductor layer extending in a first direction, fabricating a die level distribution (DLD) metallization structure, and fabricating a back end of line (BEOL) interconnect structure between the semiconductor layer and the DLD metallization structure, the BEOL interconnect structure extending in a second direction orthogonal to the first direction. Wherein fabricating the DLD metallization structure comprises fabricating an outer metallization layer extending in the first direction, the outer metallization layer comprising a metal interconnect, fabricating a first passivation layer extending in the first direction adjacent to the outer metallization layer, and fabricating a DLD metallization layer extending in the first direction. The DLD metallization layer comprises a first surface adjacent to the first passivation layer and a metal pad. The method of fabricating a semiconductor die further comprises fabricating a first via extending in the second direction orthogonal to the first direction, the first via coupling the metal pad and the metal interconnect, the first via having a first aspect ratio between 0.06-0.5, inclusively.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a side view of a die having a thick passivation layer below and above a die level distribution (DLD) metallization layer;

[0011] FIG. 2 is a cross-sectional side view of an exemplary integrated circuit (IC) package that includes a semiconductor die (“die”) having a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity;

[0012] FIG. 3 is a side view of an IC that includes a die in the IC package of FIG. 2, the die including a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity;

[0013] FIG. 4A is a side view of an embodiment of the die in FIG. 3 focusing on an exemplary DLD metallization structure which includes a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity;

[0014] FIG. 4B is a side view of an exemplary IC package including the die in FIG. 4A rotated 180° and assembled to a substrate;

[0015] FIG. 4C is a side view of another embodiment of the die in FIG. 3 focusing on an exemplary DLD metallization structure which includes a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity;

[0016] FIG. 4D is a side view of an exemplary IC package including the die in FIG. 4C rotated 180° and assembled to a substrate;

[0017] FIG. 4E is a side view of another embodiment of the die in FIG. 3 focusing on an exemplary DLD metallization structure which includes a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity;

[0018] FIG. 4F is a side view of an exemplary IC package including the die in FIG. 4C rotated 180° and assembled to a substrate;

[0019] FIG. 5 is a flowchart illustrating an exemplary fabrication process of fabricating a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. 3 and 4A-4F;

[0020] FIGS. 6A-6F is a flowchart illustrating another exemplary fabrication process of fabricating a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. 3 and 4A-4F;

[0021] FIGS. 7A-7O1 and 7N2 are exemplary fabrication stages during fabrication of the DLD metallization structure according to the fabrication process in FIGS. 6A-6F;

[0022] FIGS. 8A-8B is a flowchart illustrating an exemplary mass reflow assembly process of assembling a die to a substrate, the die including a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. 3 and 4A-4F;

[0023] FIGS. 9A-9D are exemplary assembly stages for assembling the die to the substrate according to the assembly process in FIGS. 8A-8B;

[0024] FIGS. 10A-10B is a flowchart illustrating an exemplary thermal compression assembly process of assembling a die to a substrate, the die including a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. 3 and 4A-4F;

[0025] FIGS. 11A-11D are exemplary assembly stages during assembling the die to the substrate according to the assembly process in FIGS. 10A-10B;

[0026] FIG. 12 is a block diagram of an exemplary processor-based system that can include components deployed in an IC package employing a die(s) coupled to a package substrate, wherein the die(s) can include a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. 3 and 4A-4F, according to the exemplary fabrication processes in FIG. 5 and FIGS. 6A-6F, and according to exemplary assembly processes in FIGS. 8A-8B and 10A-10B; and

[0027] FIG. 13 is a block diagram of an exemplary wireless communications device that includes radio-frequency (RF) components deployed in an IC package, wherein the IC package includes a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. 3 and 4A-4F, according to the exemplary fabrication processes in FIG. 5 and FIGS. 6A-6F, and according to exemplary assembly processes in FIGS. 8A-8B and 10A-10B.

DETAILED DESCRIPTION

[0028] With reference now to the drawing figures, several exemplary aspects of the present disclosure are described. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects. The term “adjacent” as used herein means spatially next to but not necessarily adjoining something as shown in the Figures unless specifically stated otherwise.

[0029] Aspects disclosed in the detailed description include a semiconductor die having a die level distribution (DLD) metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity. Related integrated circuit (IC) packages and fabrication methods are also disclosed. The die includes a semiconductor layer, a DLD metallization structure, and a back end of line (BEOL) interconnect structure between the semiconductor layer and the DLD metallization structure. The DLD metallization structure mechanically supports die interconnects for coupling the die to another device, such as a package substrate or another die, and redistributes signals (e.g., power, ground, information) between the die interconnects and the semiconductor layer through the BEOL interconnect structure.

[0030] In this regard, the DLD metallization structure includes a DLD metallization layer and a passivation layer disposed between the DLD metallization layer and an outer metallization layer in the BEOL interconnect structure, forming a diffusion barrier therebetween. Vias are formed in the passivation layer to couple metal pads in the DLD metallization layer to metal interconnects in a BEOL interconnect layer in the BEOL interconnect structure. It is desired to reduce the signal path resistance or, conversely, improve signal path conductivity between the die interconnects and the BEOL interconnect structure.

[0031] In this regard, in exemplary aspects, the passivation layer between the DLD metallization structure and the BEOL interconnect structure is provided as a thinner layer to minimize the distance between the DLD metallization structure and the BEOL interconnect structure. For example, the height or thickness of the passivation layer may be between 0.1 micrometer (μm) and 1 μm , inclusively, and is fabricated at this thin layer because the passivation layer's purpose is to merely act as a diffusion barrier.

[0032] In another exemplary aspect, a second passivation layer is provided between the DLD metallization structure and the die interconnects to act as a diffusion barrier between metal pads and traces in the DLD metallization structure. If the die is bonded to a package substrate as opposed to another die, for example, which requires a smooth coupling surface between dies, the second passivation layer can also be thinner. In a die-to-die bonding approach, a polishing step, such as a chemical mechanical polishing (CMP), is needed to fabricate the smooth coupling surface

and requires a thick passivation layer to include margin material which is removed when performing the polishing step. Avoiding the polishing step enables the passivation layer to be formed as a thin layer since the passivation layer does not need to include margin material and advantageously simplifies the process of fabricating the die.

[0033] Before discussing exemplary aspects starting at FIG. 2, a conventional die having a thick passivation layer which surrounds a DLD metallization layer is first discussed with reference to FIG. 1. In this regard, FIG. 1 is a side view of a die **100** having a thick passivation layer below and above a DLD metallization layer **102**. The die **100** includes a portion of a BEOL interconnect structure **104** which includes a stack of metallization layers illustrated only for simplicity as an outer metallization layer **106** and the DLD metallization layer **102**. The outer metallization layer **106** includes metal interconnects **108A-108D**. The DLD metallization layer **102** includes a metal pad **110** and metal traces **112A-112B**. The die **100** includes a thick passivation layer **114** which surrounds the DLD metallization layer **102** which utilizes copper (Cu). The thick passivation layer **114** includes a first silicon nitride (SiN) layer **116** which extends in the vertical direction (Z-axis direction) above and below the metal traces **112A-112B** and the metal pad **110** and provides a diffusion barrier to the Cu DLD metallization layer **102**. The thickness T1 in the vertical direction (Z-axis direction) of the first SiN layer **116** below the metal traces **112A-112B** and the metal pad **110** is approximately 2 micrometers (μm). The thickness T2 in the vertical direction (Z-axis direction) of the first SiN layer **116** above the metal traces **112A-112B** and metal pad **110** is approximately 0.5 μm . The thick passivation layer **114** also includes a silicon dioxide (SiO₂) layer **118** between a second SiN layer **120** and the first SiN layer **116**. The thickness T3 in the vertical direction (Z-axis direction) of the SiO₂ layer **118** between the second SiN layer **120** and the top surface of the first SiN layer **116** is approximately 4 μm . The thickness T5 in the vertical direction (Z-axis direction) of the second SiN layer **120** is approximately 0.7 μm . The die **100** also includes a polyimide layer **124** having a thickness T4 in the vertical direction (Z-axis direction) of approximately 4.5 μm . The total thickness T6 is approximately 11-15 μm .

[0034] The metal pad **110** is coupled to metal interconnects **108C** and **108D** by vias **126**. The die **100** also includes an optional die interconnect **128** which is formed in a subsequent bumping process depending on what the die **100** will be coupled to such as a substrate or another die. The die interconnect **128** couples to the metal pad **110** through a via **130**

[0035] The process in fabricating the die **100** utilizes foundry design rules to support subsequently bonding the die **100** to either a substrate or to another die. When bonding to another die in a die-die hybrid process, a top surface **132** of the metal pad **110** has to be smooth to couple to another die in a Cu to Cu hybrid bonding process. In such a process, the die interconnect **128** would not be illustrated. To smooth the top surface **132** of the metal pad **110**, a polishing process such as a CMP process is used. When using a polishing process, the passivation layer **114** has to be thicker in order to provide margin material above the DLD metallization layer **102** which is worn away during the polishing process. Additionally, the die **100** utilizes the optional polyimide layer **124** as a hard etch material to pattern vias, such as the via **130**, to metal pads, such as the metal pad **110** as opposed to using a separate photo resist which is removed after such patterning. Utilizing the polyimide layer **124** as a hard etch, some of the passivation layer **114** is compromised during the hard etching process so that additional margin material in the passivation layer **114** is required.

[0036] The thickness of the polyimide layer **124** and the thickness of the passivation layer **114** defines the height of the via **130**. As a result, the height T2+T3+T4+T5 of the via **130** is approximately 9.7 μm and the width W1 of the via **130** is approximately 20-40 μm , resulting in an aspect ratio between 2.5 and 5. The thickness T1 of the passivation layer **114** defines the height of the vias **126**. The die **100** may have been fabricated with some design rules when the DLD metallization layer **102** was made from aluminum and not copper and may be one reason why the first SiN layer **116** has the thickness T1. The height of the vias **126** in the vertical direction (Z-axis) is approximately 2 μm and a width W2 is between 1.67 μm and 4 μm yielding an aspect ratio

(height/width) between 0.5 and 1.2.

[0037] If a die need not be assembled through a die-to-die hybrid process, a polishing process during fabrication of the die can be avoided, thus, a thinner passivation layer may result, and thus shorter via heights are advantageously achieved for vias between metal pads in the DLD metallization layer and metal interconnects in the outer metallization layer. Additionally, if an optional polyimide layer is used but, in doing so, a photo resist process is utilized to pattern vias to metal pads, shorter via heights are advantageously achieved for vias between die interconnects and metal pads. Shorter via heights are proportional to higher signal path conductivity.

[0038] In this regard, FIG. 2 is a cross-sectional side view of an exemplary IC package that includes a semiconductor die (“die”) having a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity. In this example, the exemplary IC package is a three-dimensional (3D) IC (3DIC) package **200** that includes DLD metallization structures **202A-202B** including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity. The IC package **200** includes a package substrate **203** and an interposer substrate **204**. The package substrate **203** and the interposer substrate **204** commonly route signals and power and, for convenience, may both be referred to simply as a substrate **206**.

[0039] In this example, the IC package **200** includes first and second dies **208(1)**, **208(2)** that are included in respective first and second die packages **212(1)**, **212(2)** that are stacked on top of each other in the vertical direction (Z-axis direction). The first die package **212(1)** of the IC package **200** includes the first die **208(1)** coupled to the package substrate **203**. In this example, the package substrate **203** includes a first, upper and outer metallization layer **214**. The first, upper and outer metallization layer **214** provides an electrical interface for signal routing to the first die **208(1)**. The first die **208(1)** is coupled to die interconnects **218** (e.g., raised metal bumps, pillars) that are electrically coupled to metal interconnects **220** in the first, upper and outer metallization layer **214**. The first die **208(1)** includes the DLD metallization structure **202A** which couples the die interconnects **218** to the circuitry within the first die **208(1)** and includes a metal pad (not visible) to a metal interconnect (not visible) in the die **208(1)** for improved signal path conductivity. The DLD metallization structures **202A-202B** will be discussed in more detail in connection with FIGS. 3 and 4A-4F. The metal interconnects **220** in the first, upper metallization layer **214** are coupled to metal vias **222** (not visible) in the package substrate **203**, which are coupled to metal interconnects **224** in a second, bottom and outer metallization layer **216**. In this manner, the package substrate **203** provides interconnections between its first and second metallization layers **214** and **216** to provide signal routing to the first die **208(1)**. The first die **208(1)** and the second die **208(2)** include a metallization structure **202A** and **202B**, respectively, and will be discussed in more detail in connection with FIGS. 3 and 4A-4F. External interconnects **226** (e.g., ball grid array (BGA) interconnects, a.k.a. bumps) are coupled to the metal interconnects **224** in the second, bottom and outer metallization layer **216** to provide interconnections through the package substrate **203** to the first die **208(1)** through the die interconnects **218**. In this example, a first, active side **228(1)** of the first die **208(1)** is adjacent to and coupled to the package substrate **203**, and more specifically the first, upper and outer metallization layer **214** of the package substrate **203**.

[0040] In the exemplary IC package **200** in FIG. 2, an additional optional second die package **212(2)** is provided and coupled to the first die package **212(1)** to support multiple dies. For example, the first die **208(1)** in the first die package **212(1)** may include an application processor, and the second die **208(2)** may be a memory die, such as a dynamic random access memory (DRAM) die that provides memory support for the application processor. In this regard, in this example, the first die package **212(1)** also includes the interposer substrate **204** that is disposed on a package mold **230** encasing the first die **208(1)**, adjacent to a second, inactive side **228(2)** of the first die **208(1)**. The interposer substrate **204** also includes one or more metallization layers **232** that each include metal interconnects **234** to provide interconnections to the second die **208(2)** in

the second die package **212(2)**. The second die package **212(2)** is physically and electrically coupled to the first die package **212(1)** by being coupled through external interconnects **236** (e.g., solder bumps, BGA interconnects) to the interposer substrate **204**. The external interconnects **236** are coupled to the metal interconnects **234** in the interposer substrate **204** through metal vias **238** (not visible). The first die package **212(1)** includes vertical interconnects **240** to couple the second die **208(2)** to the external interconnects **226** and to the first die **208(1)** through the package substrate **203**. The second die **208(2)** also includes a DLD metallization structure **202B** which couples the external interconnects **236** to the circuitry within the second die **208(2)** and includes a metal pad and a shorter via coupling the metal pad to a metal interconnect in the second die **208(2)** for improved signal path conductivity. The DLD metallization structures **202A-202B** will be discussed in more detail in connection with FIGS. **3** and **4A-4F**.

[0041] FIG. **3** is a side view of an IC **300** that includes a die such as the die **100** of FIG. **1**, the die including a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity. A die **302** includes a BEOL interconnect structure **304** formed by a BEOL process and disposed on a front-end-of-line (FEOL) structure **306**. The FEOL structure **306** includes an active, semiconductor layer **308** that is formed on a substrate **310**. The semiconductor layer **308** extends in a first, horizontal direction, which is the X-axis and Y-axis directions as shown in FIG. **3**. The semiconductor layer **308** has a first, front side **312F** and a second, back side **312B** opposite of the first, front side **312F** in the second, vertical direction (Z-axis direction). P-type field-effect transistors (FETs) (PFETs) and N-type FETs (NFETs) **314P**, **314N** are formed in the semiconductor layer **308**. The BEOL interconnect structure **304**, as a front side interconnect structure **304**, is disposed adjacent to the front side **312F** of the semiconductor layer **308** in the second, vertical direction (Z-axis direction). The BEOL interconnect structure **304** facilitates signal routing in the die **302** on the front side **312F** of the semiconductor layer **308**. In this regard, the BEOL interconnect structure **304** includes a plurality of front side, metallization layers **316(1)-316(11)** that each include one or more metal interconnects **318(1)-318(11)** that can provide direct or indirect interconnections between the FETs **314P**, **314N** and die interconnects **320** (e.g., a solder bump) adjacent to an upper metallization layer **316(11)** of the BEOL interconnect structure **304**. The metal interconnects **318(1)-318(11)** extend in the first, horizontal direction(s) (X-and/or Y-axis directions). The BEOL interconnect structure **304** also includes via layers **322(1)-322(11)** disposed through the front side metallization layers **316(1)-316(11)** to provide interconnects between metal interconnects **318(1)-318(11)** in adjacent metallization layers **316(1)-316(11)**.

[0042] With continuing reference to FIG. **3**, a DLD metallization structure **324** includes the outer metallization layer **316(10)** extending in the first, horizontal direction (X-, Y-axes direction). The outer metallization layer **316(10)** includes metal interconnect **318(10)**. The DLD metallization structure **324** also includes a first passivation layer **326** extending in the first, horizontal direction adjacent to the outer metallization layer **316(10)**. The DLD metallization structure **324** also includes metallization layer **316(11)**, also referred to as a DLD metallization layer **328**, extending in the first, horizontal direction. The DLD metallization layer **328** is a copper layer. The DLD metallization layer **328** is the thickest layer in the die **302**, and comprises metal lines, metal traces, and metal pads to mechanically support die interconnects, electrically and physically couple die interconnects to the outer metallization layer **316(10)**, and distribute signals (power, ground, information) between die interconnects and die interconnects to metal pads in the DLD metallization layer on the periphery of the die **302**. The DLD metallization layer **328** has a first surface **330** adjacent to the first passivation layer **326** and a metal pad **332**. The DLD metallization structure **324** also includes a first via **334** extending in the second, vertical direction (Z-axis direction) orthogonal to the first direction, the first via **334** coupling the metal pad **332** and the metal interconnect **318(10)**, the first via **334** having a first height $Ht1$ in the range between $0.1\ \mu m$ and $1\ \mu m$, inclusively. The first via **334** includes a width, $Dw1$, such that the aspect ratio $Ht1/Dw1$

is between 0.06-0.5, inclusively.

[0043] The DLD metallization layer **328** has a second surface **336** opposite the first surface **330**. The DLD metallization structure **324** also includes a second passivation layer **338** extending in the first, horizontal direction adjacent to the second surface **336** of the DLD metallization layer **328**. The first passivation layer **326** and the second passivation layer **338** may consist of either SiN or SiCN without SiO₂ since its purpose is to prevent copper from diffusing into adjacent layers since no polishing process is used to make the die **302**.

[0044] The die **302** also includes the die interconnect **320** and a second via **340** extending in the second direction orthogonal to the first direction. The second via **340** couples the die interconnect **320** to the metal pad **332** and has a second height, Ht2, in the range between 0.1 μm and 1 μm , inclusively. The second via **340** includes a width, Dw2, such that the aspect ratio Ht2/Dw2 is between 0.06-0.5, inclusively. The die interconnect **320** includes a pillar **342** (e.g., Cu) and an optional solder cap **344**.

[0045] Dies can be deployed to have various DLD metallization structures. FIGS. 4A-4F address exemplary embodiments of various DLD metallization structures. Each die depicted in FIGS. 4A-4F include an FEOL structure **306** and a BEOL structure **304**. For simplicity, FIGS. 4A-4F focus on the DLD metallization structures and, thus, do not depict the FEOL structure **306** and the BEOL structure as they are shown in FIG. 3. Common elements between the die **302** in FIG. 3 and the dies in FIGS. 4A-4F are shown with common element numbers.

[0046] FIG. 4A is a side view of an embodiment of a die **400**, such as the die **302** in FIG. 3, focusing on an exemplary DLD metallization structure **402** which includes a metal pad **332** and a shorter via **334** coupling the metal pad **332** to a metal interconnect **318(10)** in the die **400** for improved signal path conductivity. The DLD metallization structure **402** includes a via **404** having a height, H2, and a width, D2, which is much greater than the height H2. The height H2 can be approximately 0.7 μm . Conductivity is proportional to the surface area of a metal interconnect and inversely proportional to the height of the metal interconnect. By utilizing the thinner second passivation layer **338**, a shorter height of the via **404** is advantageously achieved. Additionally, by utilizing the thinner second passivation layer **338**, the width of the via **404** may be opened up to this width, D2, to advantageously increase surface area **406**, and thus, the conductivity of the via **404** is increased. The width D2 can be in the range between 10 μm and 40 μm . Thus, the aspect ratio of the via **404** can be in the range 0.06 to 0.5, inclusively.

[0047] FIG. 4B is a side view of an exemplary IC package **408** including the die **400** in FIG. 4A, rotated 180° and assembled to a substrate **410**. The IC package **408** includes the die interconnect **320** soldered to the substrate **410** through a solder **412**. The IC package **408** includes an underfill material **414** to further insulate metal interconnects in the DLD metallization layer **328** and insulate the die interconnect **320** from other die interconnects.

[0048] FIG. 4C is a side view of another embodiment of a die **416**, such as the die **302** in FIG. 3, focusing on an exemplary DLD metallization structure **418** which includes a metal pad **332** and a shorter via **420** coupling the metal pad **332** to the die interconnect **320** in the die **416** for improved signal path conductivity. The die **416** also includes a polymer dielectric layer **422** (e.g. polyimide layer) to insulate the metal interconnects in the DLD metallization layer **328**. The via **420** is fabricated utilizing photo imageable material which is removed after the via **420** is formed. In other words, the polymer dielectric layer **422** is not used as a hard etch mask. In so doing, the polymer dielectric layer **422** has a thickness **424** of approximately 4.5 μm . A thickness **426** of the portion of the polymer dielectric layer **422** above a surface **427** in the second vertical direction (Z-axis direction) of the second passivation layer **338** can be in the range between 1-5 μm . The via **420** has a width D3, in a range between 10 μm and 60 μm , inclusively, and a height H3, between 1-5 μm . The DLD metallization structure **418** illustrates the via **420** where the width D3 is defined by a vertical edge **430** of the polymer dielectric layer **422**. In other words, the polymer dielectric layer **422** comprises the vertical edge **430** extending in the vertical direction (Z-axis direction) defining a

side wall **431** of the via **420**.

[0049] FIG. **4D** is a side view of an exemplary IC package **432** including the die **416** in FIG. **4C** rotated 180° and assembled to a substrate **410**. The IC package **432** includes the die interconnect **320** soldered to the substrate **410** through a solder **412**. The IC package **432** includes an underfill material **434** to further insulate the die interconnect **320** from other die interconnects.

[0050] FIG. **4E** is a side view of another embodiment of a die **436**, such as the die **302** in FIG. **3**, focusing on an exemplary DLD metallization structure **438** which includes a metal pad **332** and a shorter via **440** coupling the metal pad **332** to the die interconnect **320** in the die **436** for improved signal path conductivity. The die **436** also includes a polymer dielectric layer **422** to insulate the metal interconnects in the DLD metallization layer **328**. The via **440** has side walls in the vertical direction (Z-axis direction) and are defined by vertical edges **442** of the second passivation layer **338** and vertical edge **444** of the polymer dielectric layer **422**. The thickness **426** of the portion of the polymer dielectric layer **422** above a surface **446** in the vertical direction (Z-axis direction) of the second passivation layer **338** can be in a range below 1 μm and, preferably 0.7 μm , inclusively. The via **440** has two widths, **D4** and **D5**. The width **D4** is defined between the vertical edges **442** of the second passivation layer **338** and is between 10-40 μm , and preferably 20 μm . The width **D5** is defined between the vertical edges **444** of the polymer dielectric layer **422** and is between 20-40 μm . The height, **H4**, of the via **440** is between 1-5 μm , and preferably 3 μm . In other words, the polymer dielectric layer **422** comprises the vertical edges **444** extending in the vertical direction (Z-axis) defining first side walls **445** of the via **440**, and the second passivation layer **338** comprises the vertical edges **442** extending in the vertical direction (Z-axis) defining second side walls **447** of the via **440**.

[0051] FIG. **4F** is a side view of an exemplary IC package **448** including the die **436** in FIG. **4E** rotated 180° and assembled to a substrate **410**. The IC package **448** includes the die interconnect **320** soldered to substrate **410** through a solder **412**. The IC package **448** includes an underfill material **434** to insulate the die interconnect **320** from other die interconnects.

[0052] A DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in a die for improved signal path conductivity, including, but not limited to the exemplary DLD metallization structure **324** in FIG. **3** and the DLD metallization structures **402**, **418**, and **438** in FIGS. **4A-4F** in the related IC package **200** in FIG. **2** can be fabricated by different fabrication processes. FIG. **5** is a flowchart illustrating an exemplary fabrication process **500** of fabricating a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. **3** and **4A-4F**.

[0053] In this regard, a first exemplary step for fabricating a die with an exemplary DLD metallization structure formed in the die in the fabrication process **500** of FIG. **5** can include fabricating a semiconductor layer **308** extending in a first direction (block **502** in FIG. **5**). The next step in the fabrication process **500** can include fabricating a DLD metallization structure **324**, **402**, **418**, **438** (block **504** in FIG. **5**). The next step in the fabrication process can include fabricating a BEOL interconnect structure **304** between the semiconductor layer **308** and the DLD metallization structure **324**, **402**, **418**, **438**, the BEOL interconnect structure **304** extending in a second direction orthogonal to the first direction (block **506** in FIG. **5**). The step of fabricating the DLD metallization structure **324**, **402**, **418**, **438** further comprises blocks **508-514**. The next step of fabricating the DLD metallization structure **324**, **402**, **418**, **438** can include fabricating an outer metallization layer **316(10)** extending in the first direction, the outer metallization layer **316(10)** comprising a metal interconnect **318(10)** (block **508** in FIG. **5**). The next step of fabricating the DLD metallization structure **324**, **402**, **418**, **438** can include fabricating a first passivation layer **326** extending in the first direction adjacent to the outer metallization layer **316(10)** (block **510** in FIG. **5**). The next step in the fabrication process **500** can include fabricating a DLD metallization layer

316(11), **328** extending in the first direction, the DLD metallization layer **316(11)**, **328** comprising a first surface **330** adjacent to the first passivation layer **326** and a metal pad **332** (block **512** in FIG. 5). The next step in the fabrication process can include fabricating a first via **334** extending in the second direction orthogonal to the first direction, the first via **334** coupling the metal pad **332** and the metal interconnect **318(10)**, the first via **334** having a first aspect ratio between 0.06-0.5, inclusively (block **514** in FIG. 5).

[0054] Other fabrication processes can also be employed to fabricate a die having a DLD metallization structure with a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the DLD metallization structures **402**, **418**, and **438** in FIGS. 4A-4F in the related IC package **200** in FIG. 2. In this regard, FIGS. 6A-6F is a flowchart illustrating another exemplary fabrication process of fabricating a die having a DLD metallization structure with a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity including, but not limited to, the DLD metallization structures **402**, **418**, and **438** in FIGS. 4A-4F. FIGS. 7A-7O1 and 7N2 are exemplary fabrication stages **700A-700O1** and **700N2** during fabrication of the DLD metallization structure according to the fabrication process in FIGS. 6A-6F. The fabrication process **600** as shown in the fabrication stages **700A-700O1** and **700N2** in FIGS. 7A-7O1 and 7N2 are in reference to the cross-sectional side view of a die and, thus, will be discussed with reference to the dies **400**, **416** in FIGS. 4A and 4C which are deployed, such as the die **108(1)**, in the related IC package **200** in FIG. 2. In particular blocks **502-530** relate to the die **416** and blocks **502-526** and **532** relate to the die **400**. For simplicity, the fabrication process **600** will be focused on the DLD metallization structure fabricated therein.

[0055] In this regard, as shown in fabrication stage **700A** in FIG. 7A, an exemplary step in the fabrication process **600** is providing a die **400**, **416** with a semiconductor layer (not visible) and a BEOL structure including an outer metallization layer **316(10)** which is fabricated in the die **400**, **416** with an etch stop layer **702** (block **602** in FIG. 6A). Although not shown in the fabrication stages for simplicity, the fabrication stages address a wafer level fabrication process where the die **400**, **416** is one of many dies disposed in a wafer. The die would be fabricated in a wafer (not shown) in a wafer level fabrication process. As shown in fabrication stage **700B** in FIG. 7B, a next step in the fabrication process **600** can include depositing a first passivation layer **326** on the etch stop layer **702** (block **604** in FIG. 6A). As shown in fabrication stage **700C** in FIG. 7C, a next step in the fabrication process **600** can include patterning a photo resist material **704** over metal interconnects **706A-706D** in the outer metallization layer **316(10)** (block **606** in FIG. 6A).

[0056] As shown in fabrication stage **700D** in FIG. 7D, a next step in the fabrication process **600** can include etching through the first passivation layer **326** (block **608** in FIG. 6B). As shown in fabrication stage **700E** in FIG. 7E, a next step in the fabrication process **600** can include stripping away the photo resist material **704** (block **610** in FIG. 6B). As shown in fabrication stage **700F** in FIG. 7F, a next step in the fabrication process **600** can include depositing a barrier and seed layer **710** on a top surface **708** of the die **400**, **416** (block **612** in FIG. 6B).

[0057] As shown in fabrication stage **700G** in FIG. 7G, a next step in the fabrication process **600** can include patterning a photo resist material **712** on the barrier and seed layer **710** of the die **400**, **416** to define a form for metal interconnects **714** (block **614** in FIG. 6C). As shown in fabrication stage **700H** in FIG. 7H, a next step in the fabrication process **600** can include forming a DLD metallization layer **328** by plating Cu into the form for the metal interconnects **714** creating metal interconnects including metal pad **332** (block **616** in FIG. 6C). As shown in fabrication stage **700I** in FIG. 7I, a next step in the fabrication process **600** can include stripping away the photo resist material **712** and the barrier and seed layer **710** outside the plated Cu areas (block **618** in FIG. 6C). Please note that for simplicity the barrier and seed layer **710** in the plated Cu areas such as metal pad **332** is no longer delineated and are considered part of the metal pad **332**.

[0058] Recognizing that the die **400**, **416** will not be assembled with another die through a Cu—Cu

hybrid bonding process, a subsequent polishing step to polish the surface of the DLD metallization layer **328** will not be used. As a result, additional margin material in a passivation layer is not needed. As such, as shown in fabrication stage **700J** in FIG. **7J**, a next step in the fabrication process **600** can include depositing a second passivation layer **338** on the die **400**, **416** (block **620** in FIG. **6D**). As shown in fabrication stage **700K** in FIG. **7K**, a next step in the fabrication process **600** can include patterning and etching a photo resist material **716** to expose the second passivation layer **338** on the metal pad **332** (block **622** in FIG. **6D**). As shown in fabrication stage **700L** in FIG. **7L**, a next step in the fabrication process **600** can include etching the second passivation layer **338** to expose the surface **428** of the metal pad **332** (block **624** in FIG. **6D**). This step of the process **600** can be tailored to etch the surface **428** of the metal pad **332** so that the vertical edges of the photo resist material **716** coincide with the vertical edge of the second passivation layer **338** as shown on the die **416** in FIG. **4C**.

[0059] As shown in fabrication stage **700M** in FIG. **7M**, a next step in the fabrication process **600** can include stripping away the photo resist material **716** (block **626** in FIG. **6E**). If the die **400** is being fabricated, the process **600** proceeds to block **732** in FIG. **6F**). Continuing with fabricating the die **416**, the process **600** proceeds to block **628** in FIG. **6N1**. As shown in fabrication stage **700N1** in FIG. **7N1**, a next step in the fabrication process **600** can include applying a polymer dielectric **422** (i.e. polymer dielectric) on the die **416**, photo patterning to open the polymer dielectric layer **422** to the metal pad **332** and curing the polymer dielectric layer **422** (block **628** in FIG. **6E**). As shown in fabrication stage **700O1** in FIG. **7O1**, a next step in the fabrication process **600** can include bumping a pillar **342** and an optional solder cap **344** on the metal pad **332** (block **630** in FIG. **6E**).

[0060] Returning to block **626** in FIG. **6E**, the process **600** proceeds to block **632** of FIG. **6F** to continue fabrication of the die **400**. As shown in fabrication stage **700N2** in FIG. **7N2**, a next step in the fabrication process **600** can include bumping a pillar **342** and an optional solder cap **344** on the metal pad **332** without depositing a polymer dielectric layer (block **632** in FIG. **6F**). At the end of the wafer level fabrication process **700**, the dies disposed on a wafer including die **400**, **417** are singulated from the wafer and sent to an assembly process such as the assembly processes that will be discussed in connection with FIGS. **8-11**.

[0061] An IC package including a die having a DLD metallization structure wherein the DLD metallization structure includes a metal pad and a shorter via coupling the metal pad to a metal interconnect in a die for improved signal path conductivity, including, but not limited to the exemplary 3DIC package **200** in FIG. **2** and the DLD metallization structures **402**, **418**, and **438** in FIGS. **4A-4F** in the related IC package **200** in FIG. **2** can be assembled by different assembly processes. FIGS. **8A-8B** is a flowchart illustrating an exemplary mass reflow assembly process **800** of assembling a die to a substrate, the die including a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. **3** and **4A-4F**. FIGS. **9A-9D** are exemplary assembly stages for assembling the die to the substrate according to the assembly process **800** in FIGS. **8A-8B**. The assembly process **800** may be applied to dies with a polymer dielectric layer such as polymer dielectric layer **422** in dies **416**, and **436** but will be described in relation to the die **416**.

[0062] In this regard, as shown in assembly stage **900A** in FIG. **9A**, an exemplary step in the assembly process **800** can include aligning a die **416** (rotated 180°) with a substrate **902** having an optional bump landing pad **904** (block **802** in FIG. **8A**). The die **416** is placed on a carrier **906** to align the die **400** with the substrate **902**. The bump landing pad **904** is covered with a tacky flux **908**. The tacky flux **908** facilitates placing the die **416** on the substrate **902** with minor force.

[0063] As shown in assembly stage **900B** in FIG. **9B**, a next step in the assembly process **800** can include heating the entire attached die **416** and substrate **902** assembly in an oven (block **804** in FIG. **8A**). As shown in assembly stage **900C** in FIG. **9C**, a next step in the assembly process **800**

can include applying an underfill **910** after the tacky flux **908** has been removed (block **806** in FIG. **8B**). As shown in assembly stage **900D** in FIG. **9D**, a next step in the assembly process **800** can include curing the underfill **910** to complete IC package **912** (block **808** in FIG. **8B**).

[0064] FIGS. **10A-10B** is a flowchart illustrating an exemplary thermal compression assembly process **1000** of assembling a die to a substrate, the die including a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. **3** and **4A-4F**. The assembly process **1000** may be applied to dies with or without a polymer dielectric layer such as polymer dielectric layer **422** and will be described in relation to the die **400**.

[0065] The assembly process **1000** may be applied to the dies **400**, **416**, and **436** but will be described in relation to the die **400**.

[0066] In this regard, as shown in assembly stage **1100A** in FIG. **11A**, an exemplary step in the assembly process **1000** can include aligning a die **400** (rotated 180°) with a substrate **1102** having a solder cap **1104** (block **1002** in FIG. **10A**). The die **400** is placed on a thermal compression bonding head **1106** to align the die **400** with the substrate **1102**. The solder cap **1104** is covered with a tacky flux **1108**. The tacky flux **1108** facilitates placing the die **400** on the substrate **1102** with minor force. As shown in assembly stage **1100B** in FIG. **11B**, a next step in the assembly process **1000** can include heating the attached die **400** and substrate **1102** assembly through the thermal compression bonding head **1106** (block **1004** in FIG. **10A**). As shown in assembly stage **1100C** in FIG. **11C**, a next step in the assembly process **1000** can include applying an underfill **1110** after the tacky flux **1108** has been removed. (block **1006** in FIG. **10B**). As shown in assembly stage **1100D** in FIG. **11D**, a next step in the assembly process **1000** can include curing the underfill **1110** to complete IC package **1112**. (block **1008** in FIG. **10B**).

[0067] Electronic devices that include an IC package, wherein the IC package includes a die attached to a substrate, the die including a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. **3** and **4A-4F** in the related IC package **200** in FIG. **2**, and that can be fabricated according to, but not limited to, the exemplary fabrication processes in FIGS. **5** and **6A-6F**, and according to any aspects disclosed herein, may be provided in or integrated into any processor-based device. Examples, without limitation, include a set top box, an entertainment unit, a navigation device, a communications device, a fixed location data unit, a mobile location data unit, a global positioning system (GPS) device, a mobile phone, a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a tablet, a phablet, a server, a computer, a portable computer, a mobile computing device, laptop computer, a wearable computing device (e.g., a smart watch, a health or fitness tracker, eyewear, etc.), a desktop computer, a personal digital assistant (PDA), a monitor, a computer monitor, a television, a tuner, a radio, a satellite radio, a music player, a digital music player, a portable music player, a digital video player, a video player, a digital video disc (DVD) player, a portable digital video player, an automobile, a vehicle component, an avionics system, and a multicopter.

[0068] In this regard, FIG. **12** is a block diagram of an exemplary processor-based system that can include components deployed in an IC package employing a die(s) coupled to a package substrate, wherein the die(s) can include a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. **3** and **4A-4F**, according to the exemplary fabrication processes in FIG. **5** and FIGS. **6A-6F**, and according to exemplary assembly processes in FIGS. **8A-8B** and **10A-10B**, and according to any exemplary aspects disclosed herein. In this example, the processor-based system **1200** may be formed as an IC package **1202** such as the IC package **200** in FIG. **2**. The processor-based system **1200** includes a

central processing unit (CPU) **1208** that includes one or more processors **1210**, which may also be referred to as CPU cores or processor cores. The CPU **1208** may have cache memory **1212** coupled to the CPU **1208** for rapid access to temporarily stored data. The CPU **1208** is coupled to a system bus **1214** and can intercouple master and slave devices included in the processor-based system **1200**. As is well known, the CPU **1208** communicates with these other devices by exchanging address, control, and data information over the system bus **1214**. For example, the CPU **1208** can communicate bus transaction requests to a memory controller **1216**, as an example of a slave device. Although not illustrated in FIG. 7, multiple system buses **1214** could be provided, wherein each system bus **1214** constitutes a different fabric.

[0069] Other master and slave devices can be connected to the system bus **1214**. As illustrated in FIG. 12, these devices can include a memory system **1220** that includes the memory controller **1216** and a memory array(s) **1218**, one or more input devices **1222**, one or more output devices **1224**, one or more network interface devices **1226**, and one or more display controllers **1228**, as examples. Each of the memory system(s) **1220**, the one or more input devices **1222**, the one or more output devices **1224**, the one or more network interface devices **1226**, and the one or more display controllers **1228** can be provided in the same or different electronic devices. The input device(s) **1222** can include any type of input device, including, but not limited to, input keys, switches, voice processors, etc. The output device(s) **1224** can include any type of output device, including, but not limited to, audio, video, other visual indicators, etc. The network interface device(s) **1226** can be any device configured to allow exchange of data to and from a network **1230**. The network **1230** can be any type of network, including, but not limited to, a wired or wireless network, a private or public network, a local area network (LAN), a wireless local area network (WLAN), a wide area network (WAN), a BLUETOOTH™ network, and the Internet. The network interface device(s) **1226** can be configured to support any type of communications protocol desired.

[0070] The CPU **1208** may also be configured to access the display controller(s) **1228** over the system bus **1214** to control information sent to one or more displays **1232**. The display controller(s) **1228** sends information to the display(s) **1232** to be displayed via one or more video processor(s) **1234**, which process the information to be displayed into a format suitable for the display(s) **1232**. The display controller(s) **1228** and video processor(s) **1234** can be included as ICs in the same or different electronic devices, and in the same or different electronic devices containing the CPU **1208**, as an example. The display(s) **1232** can include any type of display, including, but not limited to, a cathode ray tube (CRT), a liquid crystal display (LCD), a plasma display, a light emitting diode (LED) display, etc.

[0071] FIG. 13 is a block diagram of an exemplary wireless communications device that includes radio-frequency (RF) components deployed in an IC package, wherein the IC package includes a DLD metallization structure including a metal pad and a shorter via coupling the metal pad to a metal interconnect in the die for improved signal path conductivity, including, but not limited to, the exemplary DLD metallization structures in FIGS. 3 and 4A-4F, according to the exemplary fabrication processes in FIG. 5 and FIGS. 6A-6F, and according to exemplary assembly processes in FIGS. 8A-8B and 10A-10B, and according to any exemplary aspects disclosed herein. The wireless communications device **1300** may include or be provided in any of the above-referenced devices, as examples. As shown in FIG. 13, the wireless communications device **1300** includes a transceiver **1304** and a data processor **1306**. The data processor **1306** may include a memory to store data and program codes. The transceiver **1304** includes a transmitter **1308** and a receiver **1310** that support bi-directional communications. In general, the wireless communications device **1300** may include any number of transmitters **1308** and/or receivers **1310** for any number of communication systems and frequency bands. All or a portion of the transceiver **1304** may be implemented on one or more analog ICs, RF ICs (RFICs), mixed-signal ICs, etc.

[0072] The transmitter **1308** or the receiver **1310** may be implemented with a super-heterodyne

architecture or a direct-conversion architecture. In the super-heterodyne architecture, a signal is frequency-converted between RF and baseband in multiple stages, for example, from RF to an intermediate frequency (IF) in one stage, and then from IF to baseband in another stage for the receiver **1310**. In the direct-conversion architecture, a signal is frequency-converted between RF and baseband in one stage. The super-heterodyne and direct-conversion architectures may use different circuit blocks and/or have different requirements. In the wireless communications device **1300** in FIG. **13**, the transmitter **1308** and the receiver **1310** are implemented with the direct-conversion architecture.

[0073] In the transmit path, the data processor **1306** processes data to be transmitted and provides I and Q analog output signals to the transmitter **1308**. In the exemplary wireless communications device **1300**, the data processor **1306** includes digital-to-analog converters (DACs) **1312(1)**, **1312(2)** for converting digital signals generated by the data processor **1306** into the I and Q analog output signals (e.g., I and Q output currents) for further processing.

[0074] Within the transmitter **1308**, lowpass filters **1314(1)**, **1314(2)** filter the I and Q analog output signals, respectively, to remove undesired signals caused by the prior digital-to-analog conversion. Amplifiers (AMPs) **1316(1)**, **1316(2)** amplify the signals from the lowpass filters **1314(1)**, **1314(2)**, respectively, and provide I and Q baseband signals. An upconverter **1318** upconverts the I and Q baseband signals with I and Q transmit (TX) local oscillator (LO) signals through mixers **1320(1)**, **1320(2)** from a TX LO signal generator **1322** to provide an upconverted signal **1324**. A filter **1326** filters the upconverted signal **1324** to remove undesired signals caused by the frequency up-conversion as well as noise in a receive frequency band. A power amplifier (PA) **1328** amplifies the upconverted signal **1324** from the filter **1326** to obtain the desired output power level and provides a transmit RF signal. The transmit RF signal is routed through a duplexer or switch **1330** and transmitted via an antenna **1332**.

[0075] In the receive path, the antenna **1332** receives signals transmitted by base stations and provides a received RF signal, which is routed through the duplexer or switch **1330** and provided to a low noise amplifier (LNA) **1334**. The duplexer or switch **1330** is designed to operate with a specific receive (RX)-to-TX duplexer frequency separation, such that RX signals are isolated from TX signals. The received RF signal is amplified by the LNA **1334** and filtered by a filter **1336** to obtain a desired RF input signal. Down-conversion mixers **1338(1)**, **1338(2)** mix the output of the filter **1336** with I and Q RX LO signals (i.e., LO_I and LO_Q) from an RX LO signal generator **1340** to generate I and Q baseband signals. The I and Q baseband signals are amplified by AMPs **1342(1)**, **1342(2)** and further filtered by lowpass filters **1344(1)**, **1344(2)** to obtain I and Q analog input signals, which are provided to the data processor **1306**. In this example, the data processor **1306** includes analog-to-digital converters (ADCs) **1346(1)**, **1346(2)** for converting the analog input signals into digital signals to be further processed by the data processor **1306**.

[0076] In the wireless communications device **1300** of FIG. **13**, the TX LO signal generator **1322** generates the I and Q TX LO signals used for frequency up-conversion, while the RX LO signal generator **1340** generates the I and Q RX LO signals used for frequency down-conversion. Each LO signal is a periodic signal with a particular fundamental frequency. A TX phase-locked loop (PLL) circuit **1348** receives timing information from the data processor **1306** and generates a control signal used to adjust the frequency and/or phase of the TX LO signals from the TX LO signal generator **1322**. Similarly, an RX PLL circuit **1350** receives timing information from the data processor **1306** and generates a control signal used to adjust the frequency and/or phase of the RX LO signals from the RX LO signal generator **1340**.

[0077] Those of skill in the art will further appreciate that the various illustrative logical blocks, modules, circuits, and algorithms described in connection with the aspects disclosed herein may be implemented as electronic hardware, instructions stored in memory or in another computer readable medium wherein any such instructions are executed by a processor or other processing device, or combinations of both. The devices and components described herein may be employed

in any circuit, hardware component, integrated circuit (IC), or IC chip, as examples. Memory disclosed herein may be any type and size of memory and may be configured to store any type of information desired. To clearly illustrate this interchangeability, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. How such functionality is implemented depends upon the particular application, design choices, and/or design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

[0078] The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a processor, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0079] The aspects disclosed herein may be embodied in hardware and in instructions that are stored in hardware, and may reside, for example, in Random Access Memory (RAM), flash memory, Read Only Memory (ROM), Electrically Programmable ROM (EPROM), Electrically Erasable Programmable ROM (EEPROM), registers, a hard disk, a removable disk, a CD-ROM, or any other form of computer readable medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a remote station. In the alternative, the processor and the storage medium may reside as discrete components in a remote station, base station, or server.

[0080] It is also noted that the operational steps described in any of the exemplary aspects herein are described to provide examples and discussion. The operations described may be performed in numerous different sequences other than the illustrated sequences. Furthermore, operations described in a single operational step may actually be performed in a number of different steps. Additionally, one or more operational steps discussed in the exemplary aspects may be combined. It is to be understood that the operational steps illustrated in the flowchart diagrams may be subject to numerous different modifications as will be readily apparent to one of skill in the art. Those of skill in the art will also understand that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0081] The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations. Thus, the disclosure is not intended to be limited to the examples and designs described herein, but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

[0082] Implementation examples are described in the following numbered clauses: [0083] 1. A semiconductor die (die), comprising: [0084] a semiconductor layer extending in a first direction; [0085] a die level distribution (DLD) metallization structure; and [0086] a back end of line (BEOL)

interconnect structure between the semiconductor layer and the DLD metallization structure, the BEOL interconnect structure extending in a second direction orthogonal to the first direction; [0087] the DLD metallization structure comprising: [0088] an outer metallization layer extending in the first direction, the outer metallization layer comprising a metal interconnect; [0089] a first passivation layer extending in the first direction adjacent to the outer metallization layer; [0090] a DLD metallization layer extending in the first direction, the DLD metallization layer comprising: [0091] a first surface adjacent to the first passivation layer; and [0092] a metal pad; and [0093] a first via extending in the second direction orthogonal to the first direction, the first via coupling the metal pad and the metal interconnect, the first via having a first aspect ratio between 0.06-0.5, inclusively. [0094] 2. The semiconductor die of clause 1, wherein [0095] the DLD metallization layer further comprises a second surface opposite the first surface; and [0096] the DLD metallization structure further comprises: [0097] a second passivation layer extending in the first direction adjacent to the second surface of the DLD metallization layer. [0098] 3. The semiconductor die of clause 2, further comprising: [0099] a first die interconnect; and [0100] a second via extending in the second direction orthogonal to the first direction, the second via coupling the first die interconnect to the metal pad, the second via having a second aspect ratio between 0.06-0.5, inclusively. [0101] 4. The semiconductor die of any of clauses 1-3, wherein the first passivation layer consists of silicon nitride (SiN). [0102] 5. The semiconductor die of any of clauses 1-4, wherein the second passivation layer consists of silicon nitride (SiN). [0103] 6. The semiconductor die of any of clauses 3-5, further comprising: [0104] a substrate, the first die interconnect coupled to the substrate; and [0105] an underfill between the second passivation layer and the substrate. [0106] 7. The semiconductor die of any of clauses 2-6, further comprising: [0107] a polymer dielectric layer extending in the first direction adjacent to the second passivation layer; [0108] a second die interconnect; and [0109] a third via extending in the second direction orthogonal to the first direction, the third via coupling the second die interconnect to the metal pad. [0110] 8. The semiconductor die of clause 7, further comprising: [0111] a substrate, the second die interconnect coupled to the substrate; and [0112] an underfill between the polymer dielectric layer and the substrate. [0113] 9. The semiconductor die of clause 7 or 8, wherein the polymer dielectric layer comprises a first edge extending in the second direction defining a side wall of the third via [0114] 10. The semiconductor die of any of clauses 7-9, wherein: [0115] the polymer dielectric layer comprises a first edge extending in the second direction defining a first side wall of the third via; and [0116] the second passivation layer comprises a second edge extending in the second direction defining a second side wall of the third via. [0117] 11. The semiconductor die of any of clauses 1-10 integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smart phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; and a multicopter. [0118] 12. A method of fabricating a semiconductor die (die) for improved signal path conductivity, comprising: [0119] fabricating a semiconductor layer extending in a first direction; [0120] fabricating a die level distribution (DLD) metallization structure; and [0121] fabricating a back end of line (BEOL) interconnect structure between the semiconductor layer and the DLD metallization structure, the BEOL interconnect structure extending in a second direction orthogonal to the first direction; [0122] wherein fabricating the DLD metallization structure comprises: [0123] fabricating an outer metallization layer extending in the first direction, the outer metallization layer comprising a metal interconnect; [0124] fabricating a first passivation layer extending in the first direction adjacent to

the outer metallization layer; [0125] fabricating a DLD metallization layer extending in the first direction, the DLD metallization layer comprising: [0126] a first surface adjacent to the first passivation layer; and [0127] a metal pad; and [0128] fabricating a first via extending in the second direction orthogonal to the first direction, the first via coupling the metal pad and the metal interconnect, the first via having a first aspect ratio between 0.06-0.5, inclusively. [0129] 13. The method of clause 12, wherein: [0130] the DLD metallization layer comprises a second surface opposite the first surface; and [0131] fabricating the DLD metallization structure further comprises: [0132] fabricating a second passivation layer extending in the first direction adjacent to the second surface of the DLD metallization layer. [0133] 14. The method of clause 13, further comprising: [0134] fabricating a first die interconnect; and [0135] fabricating a second via extending in the second direction orthogonal to the first direction, the second via coupling the first die interconnect to the metal pad, the second via having a second aspect ratio between 0.06-0.5, inclusively. [0136] 15. The method of any of clauses 12-14, wherein the first passivation layer consists of silicon nitride (SiN). [0137] 16. The method of any of clauses 12-15, wherein the second passivation layer consists of silicon nitride (SiN). [0138] 17. The method of any of clauses 14-16, further comprising: providing a substrate, the first die interconnect coupled to the substrate; and depositing an underfill between the second passivation layer and the substrate. [0139] 18. The method of any of clauses 13-17, further comprising: [0140] depositing a polymer dielectric layer extending in the first direction adjacent to the second passivation layer; [0141] fabricating a second die interconnect; and [0142] fabricating a third via extending in the second direction orthogonal to the first direction, the third via coupling the second die interconnect to the metal pad. [0143] 19. The method of clause 18, further comprising: [0144] providing a substrate, the second die interconnect coupled to the substrate; and [0145] depositing an underfill between the polymer dielectric layer and the substrate. [0146] 20. The method of clause 18 or 19, wherein the polymer dielectric layer comprises a first edge extending in the second direction defining a side wall of the third via.

Claims

1. A semiconductor die (die), comprising: a semiconductor layer extending in a first direction; a die level distribution (DLD) metallization structure; and a back end of line (BEOL) interconnect structure between the semiconductor layer and the DLD metallization structure, the BEOL interconnect structure extending in a second direction orthogonal to the first direction; the DLD metallization structure comprising: an outer metallization layer extending in the first direction, the outer metallization layer comprising a metal interconnect; a first passivation layer extending in the first direction adjacent to the outer metallization layer; a DLD metallization layer extending in the first direction, the DLD metallization layer comprising: a first surface adjacent to the first passivation layer; and a metal pad; and a first via extending in the second direction orthogonal to the first direction, the first via coupling the metal pad and the metal interconnect, the first via having a first aspect ratio between 0.06-0.5, inclusively.
2. The semiconductor die of claim 1, wherein the DLD metallization layer further comprises a second surface opposite the first surface; and the DLD metallization structure further comprises: a second passivation layer extending in the first direction adjacent to the second surface of the DLD metallization layer.
3. The semiconductor die of claim 2, further comprising: a first die interconnect; and a second via extending in the second direction orthogonal to the first direction, the second via coupling the first die interconnect to the metal pad, the second via having a second aspect ratio between 0.06-0.5, inclusively.
4. The semiconductor die of claim 1, wherein the first passivation layer consists of silicon nitride (SiN).
5. The semiconductor die of claim 2, wherein the second passivation layer consists of silicon

nitride (SiN).

6. The semiconductor die of claim 3, further comprising: a substrate, the first die interconnect coupled to the substrate; and an underfill between the second passivation layer and the substrate.
7. The semiconductor die of claim 2, further comprising: a polymer dielectric layer extending in the first direction adjacent to the second passivation layer; a second die interconnect; and a third via extending in the second direction orthogonal to the first direction, the third via coupling the second die interconnect to the metal pad.
8. The semiconductor die of claim 7, further comprising: a substrate, the second die interconnect coupled to the substrate; and an underfill between the polymer dielectric layer and the substrate.
9. The semiconductor die of claim 7, wherein the polymer dielectric layer comprises a first edge extending in the second direction defining a side wall of the third via.
10. The semiconductor die of claim 7, wherein: the polymer dielectric layer comprises a first edge extending in the second direction defining a first side wall of the third via; and the second passivation layer comprises a second edge extending in the second direction defining a second side wall of the third via.
11. The semiconductor die of claim 1 integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smart phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; and a multicopter.
12. A method of fabricating a semiconductor die (die) for improved signal path conductivity, comprising: fabricating a semiconductor layer extending in a first direction; fabricating a die level distribution (DLD) metallization structure; and fabricating a back end of line (BEOL) interconnect structure between the semiconductor layer and the DLD metallization structure, the BEOL interconnect structure extending in a second direction orthogonal to the first direction; wherein fabricating the DLD metallization structure comprises: fabricating an outer metallization layer extending in the first direction, the outer metallization layer comprising a metal interconnect; fabricating a first passivation layer extending in the first direction adjacent to the outer metallization layer; fabricating a DLD metallization layer extending in the first direction, the DLD metallization layer comprising: a first surface adjacent to the first passivation layer; and a metal pad; and fabricating a first via extending in the second direction orthogonal to the first direction, the first via coupling the metal pad and the metal interconnect, the first via having a first aspect ratio between 0.06-0.5, inclusively.
13. The method of claim 12, wherein: the DLD metallization layer comprises a second surface opposite the first surface; and fabricating the DLD metallization structure further comprises: fabricating a second passivation layer extending in the first direction adjacent to the second surface of the DLD metallization layer.
14. The method of claim 13, further comprising: fabricating a first die interconnect; and fabricating a second via extending in the second direction orthogonal to the first direction, the second via coupling the first die interconnect to the metal pad, the second via having a second aspect ratio between 0.06-0.5, inclusively.
15. The method of claim 12, wherein the first passivation layer consists of silicon nitride (SiN).
16. The method of claim 13, wherein the second passivation layer consists of silicon nitride (SiN).
17. The method of claim 14, further comprising: providing a substrate, the first die interconnect coupled to the substrate; and depositing an underfill between the second passivation layer and the substrate.

- 18.** The method of claim 13, further comprising: depositing a polymer dielectric layer extending in the first direction adjacent to the second passivation layer; fabricating a second die interconnect; and fabricating a third via extending in the second direction orthogonal to the first direction, the third via coupling the second die interconnect to the metal pad.
- 19.** The method of claim 18, further comprising: providing a substrate, the second die interconnect coupled to the substrate; and depositing an underfill between the polymer dielectric layer and the substrate.
- 20.** The method of claim 18, wherein the polymer dielectric layer comprises a first edge extending in the second direction defining a side wall of the third via.
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